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Machine Learning Approaches for Biological Age Estimation: Narrative Review of Non-Invasive and Cost-Effective Methodologies

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Biological age (BA) has proven to be a more accurate predictor of longevity and health span than chronological age, offering a dynamic measure of physiological integrity influenced by genetics, environment, and lifestyle. This narrative literature review critically examines recent advances in machine learning (ML) methodologies designed to estimate BA from minimally invasive data sources, including facial imaging, chest X-rays, brain MRIs, blood-based biomarkers, electrocardiograms, and psychological surveys. As interest grows in personalized medicine and early prevention of age-related diseases, the demand for scalable, affordable, and non-invasive BA estimation tools is increasing. Unlike previous reviews limited to a single modality, this review uniquely compares biological age (BA) prediction models based on their invasiveness, cost-effectiveness, and mean absolute error (MAE), offering a clinically oriented synthesis for non-invasive ageing diagnostics. This paper provides a needed, comprehensive, and up-to-date synthesis of these disparate approaches. Our analysis organizes these diverse methods into a unified comparative framework, systematically evaluating them on the trade-offs between predictive accuracy (MAE), invasiveness, and translational feasibility. While DNA methylation-based epigenetic clocks remain the reference standard, emerging ML-based models utilizing routine clinical or digital health data demonstrate comparable predictive performance with significantly lower cost and invasiveness. Nevertheless, emerging challenges, including variability in data quality, lack of external validation, issues of interpretability, and ethical considerations, still exist and need to be addressed in further studies. Advancing these technologies may ultimately enable the integration of non-invasive ageing biomarkers into routine healthcare, transforming risk assessment, intervention planning, and the promotion of longevity.

| **Keywords:** Machine Learning, Biological Age, Diagnostics, Biomarker, Artificial Intelligence, Personalized Medicine, Ageing

| Introduction

Ageing is characterized by the progressive decline of an organism's physiological function and increased vulnerability to death overtime (López-Otín et al., 2013). An organism as a system or part of it may age faster or slower than the "normal" rate of ageing. In humans, the rate of progressive ageing can be influenced by internal or external factors, and certain genetic alterations and lifestyle habits can accelerate, decelerate, or reverse the ageing process in time.

The concept of biological age was first introduced in 1969 and gained importance when Alex Comfort introduced the theory of the 'Test-Battery' as a method for measuring the rate at which humans age over short periods (Comfort, 1969). Then, chronological age (CA) was described as the amount of time that has passed since a person's birth, whereas the biological age reflects an individual's overall state of health and condition at a given point in their life (Ries et al., 1984). These early concepts have evolved since their introduction and today with modern methods that allow for microscopic analysis of cells. BA is most efficiently measured by epigenome studies including the methylation at certain CpG sites, such as GrimAge2 (Lu et al., 2022). Briefly, the GrimAge2 concept stems from the notion of 9 hallmarks of ageing, epigenetic alteration being one of the hallmarks (López-Otín et al., 2013) extend of introduction of altered cellular communication, genomic instability, telomere attrition, loss of proteostasis, deregulated nutrient sensing, mitochondrial dysfunction, cellular senescence, and stem cell exhaustion.

Some of these hallmarks may influence one another (e.g., cellular senescence is now known to be caused by telomere shortening but could also be caused by other factors). To date, many alternatives were developed to Horvath's method, such as those which rely on machine learning (ML) approaches, as GrimAge and other similar epigenetic tests, might be limited to use some sort of ML or statistical methods to process the large data samples. In Horvath's GrimAge2 test, the use of Cox regression with an elastic net penalty, as a model often used to assess the effect of various factors on the time until an event of interest occurs, to process data from 10,065 subjects and 13,399 samples. The data included chronological age, gender, ten epigenetic markers estimating smoking pack-years (1 pack/day for 1 year = 1 pack-year), and levels of the following biomarkers: adrenomedullin, beta-2 microglobulin, cystatin C, growth differentiation factor 15, leptin, log-transformed high-sensitivity C-reactive protein, log-transformed haemoglobin A1C, plasminogen activator inhibitor 1, and tissue inhibitor of metalloproteinases 1 (Lu et al., 2022).

Epigenetic tests, due to their high accuracy, are now the standard way for measuring BA. At first glance, they might seem ideal. They provide valuable insights into overall health and can encourage individuals to adopt healthier habits or pursue targeted treatments if the results are concerning. However, while these tests hold promise, their broader adoption and currently, their popularity is comparable to that of genealogy genetic tests and reveals certain limitations that warrant further consideration, such as cost. While not prohibitively expensive, this cost may deter some individuals if the test is not necessary. These tests are considered as invasive and require an additional effort to perform in safe manner. Even though services like myDNAge offer an easier at-home sampling experience, they aren't necessarily cheap. The process still requires users to collect a sample — either blood from a skin prick or urine — and then return the kit for analysis. This lack of true convenience remains a turn-off for many, significantly reducing their motivation to find out their biological age.

With the recent advancements in artificial intelligence (AI) and machine learning (ML), new methods arise to fill in the gaps that epigenetic tests cannot. ML algorithms can process images, written text, and numerical values and find patterns within them. This ability opens up new ways in which BA can be measured. AI can help make biological age testing more efficient, less costly, and minimally invasive, thus popularizing its assessment. Apart from that, they allow doctors to measure the BA of their patients without needing a dedicated test (like in the case of epigenetic clock tests).

With ML, factors that not only cause ageing, but also the ones influenced by it can be analysed: wrinkling of the skin, changes in blood biochemistry, senescence of brain and lung cells, or decline in heart function, just to name a few. While previous reviews have provided valuable insights into specific modalities, such as neuroimaging or blood-based biomarkers, the rapid pace of innovation across multiple disciplines necessitates a more comprehensive and updated synthesis. The primary contribution of this review is its breadth and currency. By structuring our analysis around a direct comparison of predictive accuracy (MAE), invasiveness, and accessibility, we create a practical, holistic framework for researchers and clinicians. This approach allows us to map the current state-of-the-art, identify the most promising strategies for clinical translation, and highlight the shared challenges that must be addressed to integrate BA estimation into personalized medicine.

The world's record population which in 2022 reached 8 billion people was driven predominantly by enlargement of elderly people due to life span increasing (The Lancet Healthy Longevity, 2022). The share of the oldest adults in the age structure of populations is progressively enlarged. While 20 years ago the fastest growing age group was the 80+ years, now it is the 85+ year: according to prognosis in the USA their number will double by 2036 and triple by 2049 compared to 2010 (Arias, 2010). At the same time, healthy life expectancy, which is a global indicator of population health, will not change significantly (WHO, 2016). The discrepancy between life expectancy and healthy life expectancy led to the fact that Europeans spend 20–25% of their lives in poor health (Jagger et al., 2013; Rezaei, 2024).

The well-known decline in physical capacity with aging is a focus of attention for geriatricians. Maintaining muscle strength, endurance and other parameters of physical health is crucial for preserving the functionality of older adults, ensuring their independence (Santos et al., 2016). There were elaborated a lot of questionnaires and objective tests to obtain characteristics of physical activity limitation: in a systematic review thirty-one assessment tests for mobility evaluation in elderly were identificated (Soubra et al., 2019). Various tests can assess body composition, cardiorespiratory endurance, muscular strength and endurance, and flexibility and balance separately or in combination such as The Vitality Test Battery (Mack-Inocentio et al., 2020).

Although numerous mobility assessment tools exist, evidence regarding their relative diagnostic value in functionally independent older adults with essential hypertension remains limited. This study aims to fill this gap by ranking widely used mobility tests according to their association with age and screening results for frailty, sarcopenia risk, and fall risk in middle-aged and older hypertensive patients.

| 1. Biomarkers used in biological age assessment

BA assessment relies on identifying and quantifying biomarkers that reflect the functional state of tissues, organs, and systems more accurately than CA alone. As ageing affects nearly every organ and cellular system, researchers should turn to ML approaches of determining BA instead of the costly and invasive epigenetic testing: analyzing blood samples (markers other than epigenetic), MRIs, X-Rays, images of faces and skin, as well as psychological profiling. These shift toward multi-dimensional, integrative biomarkers would mark a critical step in making this sort of diagnosis more accessible. This section focuses on how different areas of the body display accelerated ageing.

1.1 Systemic biomarkers of ageing

Systemic biomarkers of ageing, particularly those derived from blood biochemistry, offer a dynamic and integrative perspective on the molecular and physiological changes that accumulate over time. Blood-based markers reflect ongoing processes such as chronic inflammation, oxidative stress, metabolic dysregulation, and organ system decline, all key hallmarks of biological ageing. Routine clinical parameters, including levels of glucose, lipids, creatinine, C-reactive protein (CRP), albumin, and liver enzymes, have shown consistent associations with age-related morbidity and mortality, as well as offer the good accessibility and cost-effectiveness of blood tests make systemic biomarkers being scalable for ageing research and personalized health monitoring.

Ageing is marked by progressive declines in hepatic, renal, and endocrine function, which alter blood biomarker profiles. For instance, albumin levels drop due to reduced hepatic synthesis and increased capillary permeability, glucose rises reflecting insulin resistance and mitochondrial inefficiency, and urea and creatinine increase with reduced glomerular filtration rate (GFR) (Putin et al., 2016; Wang et al., 2024).

Oxidative stress amplifies these changes. Elevated γ-glutamyltransferase (GGT) and uric acid mediate 17.1% of the association between cardiovascular health (Life's Essential 8 score) and accelerated phenotypic ageing (Liu et al., 2018). With age, the amount of inflammation and immunosenescent cells increases. C-reactive protein (CRP) increases by 0.2 mg/L per decade, driven by senescent cell accumulation and inflammation. In NHANES cohorts, pro-inflammatory diets elevated white matter hyperintensity volume.

1.2 Biomarkers of facial and skin ageing

Ageing of the skin and facial structures involves molecular, structural, and functional changes driven by mitochondrial dysfunction, oxidative stress, hormonal shifts, and cumulative DNA damage. These processes manifest as wrinkles, volume loss, pigmentation irregularities, and altered facial contours. The human face and skin offer a visually accessible and biologically informative source of ageing biomarkers, making them increasingly valuable in the development of machine learning models aimed at estimating biological age. Facial morphology, skin texture, wrinkle formation, pigmentation changes, and loss of

elasticity are among the most recognizable hallmarks of ageing and reflect underlying physiological and molecular alterations, including collagen degradation, oxidative stress, and reduced cellular turnover. Advances in computer vision and deep learning have enabled the quantification of subtle facial features and skin attributes from digital images, allowing for the construction of predictive models that can estimate biological or perceived age with high accuracy. These models not only capture visible signs of chronological ageing but also integrate cues associated with lifestyle, environmental exposures, and systemic health, positioning the face and skin as non-invasive, real-world biomarkers with growing clinical and consumer relevance.

Intrinsic ageing arises from genetically programmed processes that impair cellular repair and regeneration. Over time, fibroblasts in the dermis produce less collagen and elastin, leading to reduced skin elasticity and tensile strength (Guinot et al., 2002). The decline in collagen synthesis is exacerbated by the accumulation of senescent cells, which secrete pro-inflammatory cytokines that degrade the extracellular matrix (ECM) (P. Chevalier et al., 2020). Hormonal changes further modulate ageing trajectories. Postmenopausal estrogen depletion reduces skin thickness and hydration, while androgens in males contribute to sebaceous gland hyperactivity and thicker, more resilient skin (Keaney et al., 2016).

Extrinsic ageing is predominantly driven by ultraviolet (UV) radiation, which accounts for up to 80% of visible facial ageing. UVB rays penetrate the epidermis, inducing DNA crosslinks and thymine dimer formation. UVA rays, however, generate reactive oxygen species (ROS) in the dermis, activating matrix metalloproteinases that degrade collagen (Krutmann et al., 2003). Chronic sun exposure also disrupts melanocyte function, leading to hyperpigmentation and lentigines. Smoking compounds these effects by constricting dermal blood vessels, reducing oxygen delivery, and inhibiting collagen synthesis (Guinot et al., 2002).

Over time, structural changes also arise. The medial cheek fat pad, which provides midface support, diminishes with age, leading to nasolabial fold prominence and jowl formation (Bravo et al., 2020) while repetitive facial expressions induce wrinkles that become permanent over time (Rohrich et al., 2021).

1.3 Biomarkers of the cardio-respiratory system ageing

The cardio-respiratory system undergoes significant age-related remodelling, offering valuable biomarkers for the estimation of biological age. In the respiratory system, ageing induces both structural and functional changes, which can all be visualized through subtle radiographic alterations in chest imaging. The cardiovascular system also exhibits hallmark features of ageing. The electrocardiogram (ECG), a widely available and non-invasive tool, reflects many of these age-related cardiac alterations.

The aortic knob widens laterally, with calcifications frequently outlining the arch. In longitudinal studies, 79% of healthy men exhibited aortic enlargement over 16 years (independent of hypertension). While the cardiac silhouette may slightly enlarge due to pericardial fat accumulation, the cardiothoracic ratio rarely exceeds 0.5 in healthy ageing.

Flattened diaphragms and increased retrosternal airspace are common, reflecting loss of elastic recoil. Hyperinflation prevalence doubles from 27% to 54% between the sixth and ninth decades, even in asymptomatic individuals. Costophrenic angle blunting occurs in 30% of individuals over 70, related to subpleural fibrosis rather than prior inflammation (Ensor et al., 1983).

Fibrosis (scarring) of the sinoatrial node and His-Purkinje system prolongs P-wave and QRS durations. Interatrial conduction block, present in 23% of elderly atrial flutter patients, correlates with a 175 ms P-wave duration, 16 ms longer than in younger cohorts (Ślimak et al., 2022).

A study on guinea pigs across their lifespan revealed that sinoatrial node (SAN) dysfunction with age is linked to a progressive loss of connexin 43 (Cx43) protein (Jones et al., 2004). This manifests as QT interval prolongation, which increases by 10 ms per decade in adults without hypertension (Zhang et al., 2021).

Resting heart rate also tends to rise with age, especially in those with schizophrenia, partly due to antipsychotic-induced sodium channel blockade (Blom et al., 2014). Structurally, changes like prolonged P-waves suggest right atrial enlargement, while fragmented QRS complexes are signs of conduction issues common in older adults with increased heart mass (Ślimak et al., 2022; Piccirillo et al., 2022).

1.4 Biomarkers of neurological and psychological ageing

Ageing induces a complex of structural, metabolic, and molecular changes in the brain, which manifest as distinct neuroimaging and electrophysiological signatures. It is accompanied by psychological and cognitive transformations that reflect both neurobiological changes and psychosocial adaptations.

Progressive cortical thinning and subcortical volume loss represent hallmarks of brain ageing. The hippocampus exhibits an annual atrophy rate of 1–2% after age 50, correlating with episodic memory decline. Expansion of the lateral ventricles by 2–3% per decade reflects parenchymal volume loss and disrupted cerebrospinal fluid dynamics (Dartora et al., 2024).

Quantitative susceptibility mapping reveals age-related iron accumulation in the substantia nigra and subthalamic nucleus, increasing by 0.8–1.2 ppb/year (Falahati et al., 2024). This iron dysregulation contributes to oxidative stress, accelerating neuronal loss in Parkinson's disease.

White matter hyperintensities increase by 1.5–3.0 cm^3 per decade, reflecting small vessel disease and blood-brain barrier leakage (Melo Van Lent et al., 2023).

Recently, studies have shown that microplastics (MPs) introduce many complications, including accelerated ageing. MPs induce chronic activation of microglia, leading to elevated interleukin-6 and tumour necrosis factor-alpha levels, as demonstrated in rodent models of global cerebral ischemia (Kim et al., 2025). Prolonged inflammation accelerates neuronal death and synaptic loss, processes detectable via MRI as reduced grey matter volume and increased white matter hyperintensities.

Subjective age — the discrepancy between felt and chronological age — serves as a robust predictor of health outcomes. Individuals who perceive themselves as older than their actual age exhibit higher all-cause mortality risk and accelerated epigenetic ageing (Zhavoronkov et al., 2020; Stephan et al., 2015). For example, rheumatoid arthritis patients report 18% older subjective ages than healthy peers, mediated by chronic pain and mobility limitations (Schönstein et al., 2020). Conversely, centenarians in Sardinia demonstrate psychological resilience linked to social desirability and community engagement, which buffer against age-related stress (Fastame MC, 2020). Observed deviations between predicted brain age and chronological age, defined as the brain age gap, are gaining increasing recognition as a clinically relevant index of neurological health and biological ageing.

Self-reflection and lifelong learning build cognitive reserve, delaying dementia onset by 5–7 years. Older adults with high reserve maintain intact memory despite amyloid positivity, likely through enhanced default mode network connectivity (Bartrés-Faz et al., 2023).

Alzheimer's disease (AD), the most common form of dementia, can be diagnosed and is a strong marker for BA. It is characterized by amyloid-β plaques, neurofibrillary tau tangles, and synaptic loss, which correlate with progressive cognitive decline. Epigenetic ageing accelerates in AD patients and shows strong associations with neuritic plaque burden and amyloid load (Thrush et al., 2022; Levine et al., 2015). Vascular contributions to dementia, particularly cerebral small vessel disease, further complicate the ageing-neurodegeneration link (Inoue et al., 2023).

| 2. Predicting biological ageing with machine learning

Ageing involves numerous factors, causing quantifiable and previously unquantifiable changes throughout the body. ML and AI are now enabling us to measure these age-related deteriorations, particularly in tissues that can be photographed or scanned. This section will explore various cutting-edge ML models that leverage different biological data sources to predict biological age.

2.1 Face image analysis

The look of a face is one of the most prominent markers of how well an individual age. It's how humans tend to assess BA every day (e.g., someone ages well when their face doesn't change much over a long period).

PhotoAgeClock (Bobrov et al., 2018) primarily focuses on predicting CA, not BA. It does so use high-resolution images of human skin around the eyes. The image and label data used to train the model were sourced from public facial image datasets: FG-NET, MORPH, Adience, and IMDb-WIKI. To benchmark the model's performance against other existing methods of determining BA (like the already mentioned GrimAge DNA methylation test and blood biochemistry age estimators).

Using eye-corner images, the team trained Xception NNs and then added skip connections to increase accuracy. The study sought to determine how important the eye-corner area is for making an accurate age prediction. Wrinkles, pigmentation, and fine skin details around the eye corner were hypothesized to be strong indicators of

ageing. Portions of the image were gradually blacked out or replaced with zero tensors, beginning with small areas (e.g., just the eye) and progressing to larger areas (e.g., half the image). The occluded images were passed through the trained NN to observe changes in prediction error. Accuracy dropped significantly when key eye areas were occluded, especially for older individuals, signifying a correlation to CA.

Overall, PhotoAgeClock can predict chronological age with high accuracy. The study displayed a mean absolute error (MAE) of 2.38 years for Xception without skip connections and 2.30 years with skip connections. The test 'outperformed' a DNA-methylation model (MAE = 2.70); however, that may be because the DNAm model was trained to predict BA, and the results of the models were validated on CA.

Another model, FaceAge, sourced its training data from the IMDb-Wiki database, which included 174,195 labelled images (Dennis B, 2025). The researchers filtered out the unusable data, leaving them with a dataset of 57,304 people. The dataset for ages 60+ underwent manual curation to ensure image quality and reduce noise in labels, given the clinical relevance of this age group. The team compared the results of FaceAge to data derived from the UTK dataset.

Unlike PhotoAgeClock, FaceAge focuses on estimating BA, prioritizing clinical relevance over strict chronological accuracy. The pipeline uses face localization through MTCNN and age estimation via an adapted Inception-ResNet v1 architecture. By emphasizing older age groups (60+ years) and avoiding overfitting during fine-tuning, the model aligns with its goal of predicting BA processes, which are more relevant to clinical oncology outcomes. While capable of predicting chronological age, the model's design is tailored to capture markers of BA.

This is reflected in the results. For the entire dataset, MAE = 5.87 years; however, in the curated dataset for the 60+ age range (which was prioritized): 4.09 years. Fine-tuned models showed lower MAE (<3 years) for healthy controls but reduced prognostic power in clinical datasets.

2.2 Chest X-ray analysis

X-ray imagery gives insight into the internal structures of the body of an individual. This is useful for BA prediction, as not all health features are visible on the outside. The chest area is highly potent in degrading over time (Hockhegger et al., 2012). Hence, using chest radiography as a parameter for determining BA could be a promising method.

Raghu V.K. et al. developed CXR-Age, a deep learning model designed to estimate biological age from chest X-rays (CXR) (Raghu et al., 2021). The model was trained on data from 116,035 individuals and validated on two separate testing sets: 75% of the CXR dataset of 40,967 from the PLCO (Prostate, Lung, Colorectal, and Ovarian Cancer Screening Trial) and the CXR dataset of 5,414 from the NLST (National Lung Screening Trial).

The development of CXR-Age consisted of two stages. In Stage 1, a pre-training model was provided with a CXR image to give an estimate of chronological age. The Stage 1 model was the starting point for a second model trained to estimate BA from a CXR image. To train the model, it was necessary to assign a BA label to each training dataset participant, as opposed to the CA label linked to each participant in the Stage 1 model. Biological age labels were calculated using actuarial life tables and survival models. For those who had died, labels were based on differences between expected and actual age at death, while for others, time-to-death was estimated using statistical models. These labels were solely for model training and were not reported in the results.

Overall, CXR-Age outperformed chronological age and traditional radiographic findings in predicting all-cause and cardiovascular mortality. Most individuals had a CXR-Age within five years of their chronological age (PLCO: 73%; NLST: 68%). However, a subset showed significant deviation, with CXR-Age at least five years older or younger, more prevalent in NLST due to heavier smoking histories.

2.3 Brain MRI analysis

The degradation of the brain over time is likely the most detrimental effect of ageing. It is the primary risk factor for Alzheimer's disease, memory loss, and other neurodegenerative disorders (Yankner et al., 2008). Analyzing brain MRIs would not only help estimate the BA of the whole body but also assess the BA of the brain with greater accuracy (which is likely more important).

Age-Net is a deep learning (DL) model that analyses brain MRIs to predict the age of a subject (Armanious et al., 2021). The model used the open-source IXI dataset of brain MRI scans from 562 healthy subjects. From the 562 scans, 420 were used for training, 92 for validation, and 50 for testing.

A custom deep convolutional neural network (CNN) architecture was used. It's a hybrid design incorporating Inception v1 and SqueezeNet architectures. Two input strategies were tested: 1) feeding the entire 3D GM volume to the network, and 2) dividing the volume into

smaller 3D chunks and feeding each chunk separately. Gender information was also incorporated in some models.

Age-Net was compared to other existing methods for brain age estimation using MRI: 2D-Huang (MAE = 3.529), 3D-Ueda (MAE = 3.705), and 3D-Peng (MAE = 2.741). The team determined the MAE for each of the tested models: Age-Net-Volume (MAE = 2.658), Age-Net-Chunk (MAE = 2.283), Age-Net-Gender (MAE = 1.955).

2.4 Blood biomarkers analysis

Mamoshina et al. developed a blood marker-based model to determine the age of an individual (Mamoshina et al., 2018). The team used four datasets containing 20,699 samples for the Canadian population, 65,760 samples for the South Korean population, and 55,920 samples for the Eastern European population. Each sample contained, 15 biochemistry markers, including albumin, glucose, haemoglobin, cholesterol, sodium, urea, LDL cholesterol, triglycerides, hematocrit, HDL cholesterol, total protein, calcium, creatinine, potassium, and total bilirubin, and four cell count markers, including erythrocytes, and platelet count (19 features in total). The data was split into an 80/20 train-test ratio.

The model takes a vector of blood test values and returns a single value of patient age. The performance of three separate DNNs were trained and tested on four distinctive datasets: The South Korean, Canadian, Eastern European, and the combined dataset (all three) with either 19, 20, or 21 input features.

For the Canadian-trained model, the MAE validated on Canadian patients was 6.36 years. For the South Korean-trained model, the MAE validated on S. Korean patients was 5.59 years. For the Eastern-European-trained model, the MAE was validated on E. European patients were 6.25 years. When the model trained on the combined datasets was compared to the combined populations, the MAE was 5.94 years; however, when the same model was compared to an independent NHANES (National Health and Nutrition Examination Survey) dataset (so US patients), the MAE was 9.93 years. The Canadian-trained model performed even worse with the NHANES test, with an MAE of 12.65.

Another study developed by Sagers L. et. al. also makes an age prediction from lab data (Sagers et al., 2020). The study aimed to predict chronological age using 356 blood biomarkers (e.g., serum glucose, creatinine) across 67,563 individuals. Data were obtained from the CDC NHANES datasets spanning the years 1999-2016. This dataset includes a representative sample of the non-institutionalized U.S. population.

The team used the random forest technique to train their models. Models trained on one age group generally performed poorly when applied to others, demonstrating the need for age-specific models. Hence, separate models were trained and validated for four age groups.

The study identified nonlinear trajectories for 28.4% of analytes, with significant slope changes around the 16-17-year mark, suggesting a critical biological transition in late adolescence. The four models trained on the four age groups displayed the following MAEs: 0.87 years (1-18), 5.15 years (18-45), 2.20 years (45-65), and 4.30 years (65+).

Mamoshina et al. and Sagers et al. differ in their approaches to age prediction from blood markers. Mamoshina's DNNs, using 19 features across diverse populations, struggle with cross-population applicability. Sagers' random forest models, with 356 biomarkers, achieve better accuracy using age-specific models, highlighting nonlinear trajectories around late adolescence.

2.5 ECG analysis

Heart activity is impacted by age (Craft et al., 1995). Older individuals exhibit slower heart rates, reduced heart rate variability (HRV), as indicated by smaller deviations in R-R intervals, slower electrical signal transmission from the atria to the ventricles, and prolonged signal conduction times, just to name a few.

One of the most insightful markers for heart activity is the ECG. Baek Y. et. al. developed an AI model that sought to analyze this marker to predict the mortality of a given patient. ECG-heart age was estimated using standard 12-lead ECGs (Baek et al., 2023). Data from 522,109 ECGs from a sample of 226,476 adults aged over 18 years was used. An 80/20 train-test split was used.

The deep neural network, incorporating multi-head attention and bi-directional LSTM blocks, processed data from eight ECG leads (I, II, V1–V6) sampled at 500 Hz over 10 seconds, excluding derived leads (III, aVR, aVL, aVF). The study evaluated biological heart age predictions against various outcomes, including all-cause mortality, cardiovascular-related mortality, major adverse cardiovascular events (MACE), and cardiovascular hospitalizations, based on ECG features such as PR interval, QRS duration, and QT interval.

A holdout test of 97,058 ECGs (34,317 patients) was used to validate the model. The team reported the heart age prediction of their model to have an MAE of 5.8 ± 3.9 years.

2.6 Psychological and cognitive analysis

Psychological and cognitive function-based tests are not as developed yet as other ways of determining BA, however, they deserve further investigation due to their particularly non-invasive nature (Erema et al., 2022). There are a few ways of testing for BA using psychological and cognitive markers. The most common category would rely on marking behavioral features (this would be similar to tests for dyslexia).

Zhavoronkov A. et. al. developed PsychoAge and SubjAge in 2020 to predict chronological age based on these parameters in a cohort of mentally healthy individuals (Zhavoronkov et al., 2020). PsychoAge was trained to approximate chronological age, and SubjAge – subjective or biological age. The models were trained using data from the MIDUS study (6071 individuals in total). 50 psychosocial features were selected as input variables for the models. These features were categorized based on their MIDUS descriptions, including aspects of health, relationships, life changes, occupation, community involvement, personality, psychological beliefs, demographics, and well-being.

Deep neural networks (DNNs) were used for both PsychoAge and SubjAge. Five-fold cross-validation was employed during training. Cross-validation helps to assess the model's ability to generalize to unseen data and prevent overfitting.

In the end, the models achieved an MAE of 6.7 years for PsychoAge and 7.3 years for SubjAge. The subsequent feature importance analysis allowed for the ranking of the features and the selection of the most potent markers of ageing among them. For both models, sex life, vigorous activity (like being out of breath), taking prescription blood pressure medication, and marital status were the most important features, both physically and mentally.

A schematic representation of various cutting-edge ML models that leverage different biological data sources to predict biological age is shown in Figure 1.

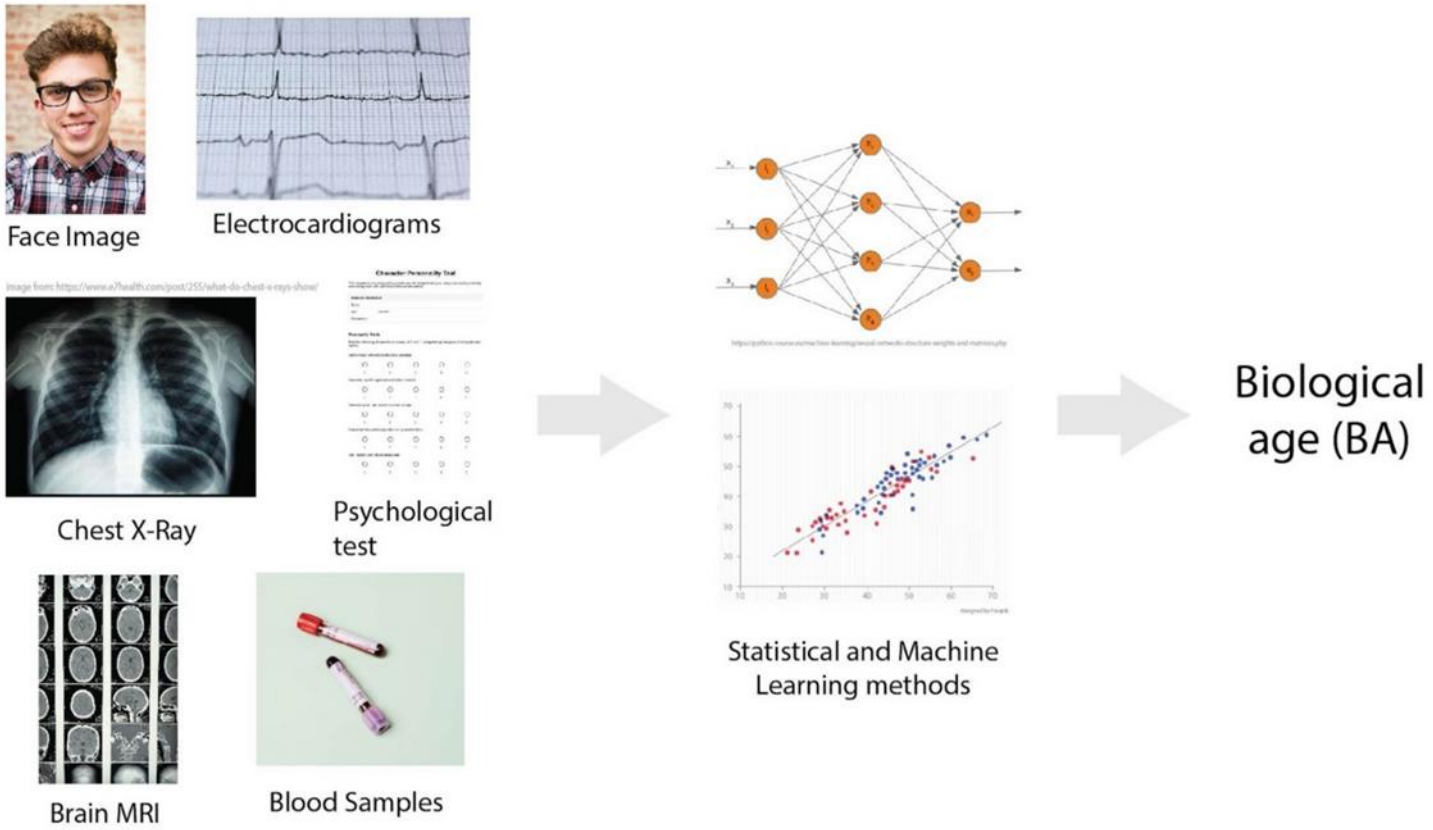

Figure 1. Various ML models that leverage different biological data sources to predict biological age.

| 3. Evaluation of the models

The models presented above are primarily designed to determine an individual's age from data that is either easily accessible or obtained while testing for something else. While some models display low MAE, others do not. The survey in Section 2 reveals a consistent, defining pattern: a fundamental trade-off between predictive accuracy, invasiveness, and accessibility. No single method is superior on all fronts. When assessing the accuracy of each model, there are a few things to consider:

First, the purpose of the model. There have been multiple attempts at determining BA, which is a lot harder to estimate than CA. While the majority of the models presented do predict BA or have a BA-predicting counterpart, some do not. For now, CA models tend to be more accurate simply because data with truthful labels is easier to obtain. The exception is the most accurate model — Age-Net, which uses brain scans from which a lot more data can be extracted per subject than, for example, from a face image. It is the only model presented that surpassed the 2-year MAE; however, this precision comes at the highest cost and lowest scalability, restricting its utility. On the other end of the spectrum lie the most scalable and accessible methods. Models using facial images (FaceAge) or psychological surveys (PsychoAge) offer unparalleled ease of deployment but with currently lower accuracy (MAE > 4 years).

Some models just use markers whose changes have a lower correlation to the ageing of a subject. Changes in the psychological markers and ECGs, for example, may very likely have a weaker correlation to the BA of a subject when compared to, e.g., brain scans or face images.

The quality and often complexity of the model also contribute to its accuracy. This may be influenced by the novelty of the particular markers. Newer, less developed markers will likely have lower-quality models. This may be the cause for the low accuracy in PsychoAge and SubjAge.

One of the most challenging obstacles to overcome with BA prediction is acquiring large amounts of accurate training and testing data. Most of the models presented either have a lot of mediocre data, sourced from open-source databases, solely to gather as much data as possible, or little, very high-quality, accurate data, which was often manually curated. For a valid prediction, the model should be trained on a lot of high-quality data, which is difficult to acquire, especially for BA prediction.

For accurate BA prediction models, data points need to be labelled with the biological ages of the subjects. There are a few ways of achieving this:

Using the time of natural death of a subject to determine their BA, as done by e.g., Fekrazad A., where the data was taken from deceased individuals whose images were found on the internet [Fekrazad, 2023]. This approach, however, requires all the subjects to be deceased, which would either yield data points with insufficient parameters from the subjects (when data was sourced from past studies) or only data on older individuals who are not fully representative and wouldn't produce a good model for younger age groups. Raghu et. al. used a hybrid with CXR-Age (Raghu et al., 2021). In this study, the researchers defined BA as the age at which an individual's risk of mortality matched the risk of an average person of that age, according to actuarial life tables. For participants who died during follow-up, their BA was calculated using Figure 2.

$$BA = CA\ at\ time\ of\ measurement + (expected\ age\ at\ death - CA\ at\ death)$$

Figure 2. Biological age is calculated by adding the difference between the expected age at death and actual chronological age at death to the chronological age at the time of measurement (and estimation of predicted age at death) Adapted from (Raghu et al., 2021).

The expected age at death was derived from 2015 actuarial life tables published by the U.S. Social Security Administration. So, if an individual died exactly and their expected age of death (expected age at death = CA at death), their BA is the same as their CA.

For participants who did not die during follow-up, their BA was estimated using sex-specific Gaussian survival models. These models 1) were trained on data from the control arm of the PLCO trial; 2) predicted each participant's time-to-death based on their sex and other survival-related variables; 3) used the newly predicted age at death for the CA at death in the formula above.

Armanious et. al addressed the lack of "ground-truth labels" by developing an iterative data-cleaning strategy to find correct BA labels in a dataset with CA labels (Armanious et al., 2021). The process aimed to refine a "typical-ageing dataset" where BA ≈ CA by systematically removing outliers. The researchers used the Age-Net-

Gender framework (proven effective for CA estimation) as the baseline model. For each patient, a threshold BA was calculated to flag outliers exhibiting accelerated ageing (BA > CA) or delayed ageing (BA < CA). Patients falling outside the typical ageing range were flagged as outliers and removed. The process was repeated with a reshuffled dataset, and the model was retrained from the beginning. To prevent removing typically ageing individuals, only those flagged in multiple iterations were excluded. Iterations continued until no outliers were detected in three consecutive runs, and all atypically ageing patients were assumed to be removed. Hence, the model was trained on individuals whose CA was very similar to their BA, so their calendar ages could be used as labels. This could yield inaccurate results for individuals whose BA deviates a lot from their CA, as the models were not trained on such cases.

The method of validation also differs from one model to another. The models use different data as a validation set. For models that predict BA, the labels in the validation set were often derived in the same way as the training set, leaving room for error in the label prediction process itself. It can have a decent inaccuracy without being detected. Hence, establishing labels cheaply and easily remains the greatest obstacle in determining BA with ML and AI.

A summary of all models mentioned in Section 3 is presented in Table 1.

Table 1. Summary of all presented models. Some estimate the chronological age (CA), while others – the biological age (BA).

Model	Input type	Best MAE (years)	Purpose (BA/CA)	Key Insights
PhotoAgeClock (Bobrov et al., 2018)	Eye corner images	2.30 years (with skip connections)	CA	High accuracy for CA prediction; focuses on fine skin details like wrinkles and pigmentation.
FaceAge (Dennis, 2025)	Whole-face images	5.87 years (entire dataset), 4.09 years (60+ curated dataset)	BA	Prioritizes BA prediction for clinical relevance, especially for the older age group.
CXR-Age (Raghu et al., 2021)	Chest X-rays	Not directly reported (73% predictions within 5 years of CA)	BA	Strong in predicting all-cause and cardiovascular mortality, leveraging deviations from CA.
Age-Net (Armanious, 2021)	Brain MRIs	1.96 years (Age-Net-Gender)	BA	Highly accurate in BA prediction, outperforming previous MRI-based methods.
Mamoshina et al., 2018	Blood biomarkers	5.59 years (S. Korean model on S. Korean population)	CA	Performance varies significantly by population, suggesting strong population specificity.
Sagers et. al., 2020	Blood biomarkers (356)	2.20 years (45-65 age group)	CA	Requires age-specific models for optimal performance. Nonlinear trajectories of biomarkers provide deeper biological insight.
Baek et al., 2023	12-lead ECGs	5.8 ± 3.9 years	BA	Strong in mortality prediction; aligns ECG markers with BA indicators.
PsychoAge (Zhavoronkov et al., 2020)	Psychosocial data	6.7 years	CA	Relatively high MAE but offers non-invasive BA estimation.
SubjAge (Zhavoronkov et al., 2020)	Psychosocial data	7.3 years	BA	Focuses on BA and mental health indicators.

| Conclusions

Promoting the measurement of BA is important due to its significant health benefits in enhancing the overall well-being of a population or individual. As one of the most reliable indicators of overall health, a BA lower than CA suggests opportunities for individuals to intervene more effectively and can even serve as motivation to take proactive health measures.

CA predictors discussed in this article lack direct medical applications but can be useful for statistical purposes, such as recording the age groups frequenting certain locations with tools like PhotoAgeClock.

The biggest challenge in determining BA is assigning accurate BA labels to individuals cost-effectively and efficiently. Most of the studies reviewed rely on workarounds such as statistical models, probabilities, and outlier elimination. However, there remains a significant shortage of high-quality, reliable, and validated BA data to serve as robust labels for these models. The most accurate BA prediction model identified was Age-Net, achieving an MAE of less than two years. It might even prove useful in some medical scenarios (for example, as a preliminary age test). Discussing the least accurate models is less meaningful, as new, less effective models may continually emerge. Studies that trained multiple models, one for each age group, often demonstrated higher accuracy when tested on the respective age groups than the corresponding general model. This demonstrates a need for multiple models.

| Future perspectives

In making models less invasive, privacy concerns must be addressed. For instance, models accessible via a web app that requires users to share potentially sensitive information with the app can raise privacy issues, particularly as public awareness of data security grows. Some of the models discussed are open source, allowing them to be downloaded, run locally without server dependence, and even modified or improved by others as enhancing an existing model may often be more practical than creating one from scratch. For example, models such as FaceAge or ECG-Age may soon aid in clinical triage or preventative screenings by highlighting patients whose BA significantly deviates from their CA, prompting further evaluation.

The comprehensive, cross-modal analysis conducted in this review not only maps the current state of the art but also illuminates the most critical directions for future research. Based on our comparison of accuracy, invasiveness, and accessibility, an ideal BA model must:

1. use a marker that highly correlates with BA;
2. be trained on data with real or highly accurate BA labels;
3. consist of age-group-specific models to improve accuracy;
4. use validation data with accurate BA labels, ideally derived from an independent method if the training data's accuracy is uncertain;
5. be open source to potentially address privacy concerns and encourage collaborative improvement;
6. if released via a website, employ proper data safety precautions and anonymize users.

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| Conflicts of Interest

The authors declare no conflicts of interest.

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